

# Retrieval-Augmented Generation for Hallucination Mitigation in Large Language Models: A Systematic Review of Architectures, Evaluation and Open Problems

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**Abstract**—Large language models generate fluent text that is not reliably supported by any verifiable source, and the resulting hallucinations are the principal obstacle to deploying them where correctness carries cost. Retrieval-Augmented Generation (RAG) has become the dominant architectural response, grounding generation in retrieved evidence and supplying the provenance that purely parametric models cannot. This paper systematically reviews 35 primary studies published between 2019 and 2026, drawn from IEEE, ACM, Springer, Elsevier, MDPI, Wiley, the ACL Anthology and arXiv, with every bibliographic record resolved programmatically against publisher metadata. We trace the evolution of the field across four shifts: the externalisation of knowledge from parameters to an index; the scaling of retrieval infrastructure; the transition of retrieval from a fixed pipeline stage to a decision made during generation; and the entry of verification into the generation loop. We compare mitigation architectures along retrieval trigger, verification mechanism, training requirement and evaluation protocol, and argue that the cross-paper numeric comparison commonly presented in this literature is methodologically invalid because reported systems differ in base model, retriever and evaluation split. Mining limitations across the corpus yields a frequency distribution dominated by generalization (24.8%), evaluation validity (17.1%) and computational cost (13.3%), from which we derive seven evidence-grounded research gaps. The most consequential is that grounding is assumed rather than verified: recent word-level annotation shows RAG systems hallucinate even when correct evidence has been retrieved, which undercuts the paradigm's central assumption. We propose seven future directions and three novel research ideas, including entailment-gated generation and semantic-entropy triggered retrieval. We conclude that, given formal results establishing that hallucination cannot be eliminated, the field's objective should be restated as bounded and measurable residual risk rather than elimination.

**Index Terms**—Retrieval-augmented generation, large language models, hallucination, factuality, grounding, attribution, information retrieval, natural language generation, systematic review

## I. INTRODUCTION

Large language models produce text that is fluent, well-structured and frequently wrong in ways that leave no surface trace. This failure mode - commonly termed hallucination - is not an incidental defect to be engineered away but a consequence of how these systems are built and evaluated. [1] argues formally that hallucination is an innate limitation of the model class

and cannot be eliminated by scale, data quality or guardrails. [2] argues that prevailing benchmark scoring compounds the problem by rewarding a confident guess over an admission of ignorance, making fabrication the statistically optimal response. Taken together, these results reframe the engineering objective: the question is not how to remove hallucination but how to bound it, measure it, and make its residual level visible to whoever depends on the output.

The empirical case for external knowledge predates that theoretical framing. [3] showed that pre-trained language models recall a substantial amount of relational knowledge without supervision, but that recall is uneven and highly sensitive to how a query is phrased. [4] sharpened this into an operational finding: parametric memory performs well on popular entities and degrades sharply on the long tail, exactly where an external source would help most. Parametric knowledge is therefore real, unreliable, unattributable and un-updatable without retraining - four properties that make it unsuitable as the sole knowledge substrate for a deployed system.

Retrieval-Augmented Generation was introduced to address all four. [5] combined a dense retriever with a generative reader trained end to end, demonstrating that conditioning generation on retrieved passages improves factual specificity and, critically, allows the knowledge source to be replaced without retraining the model. [6] supplied the dense retriever that made this practical, and [7] showed retrieval could be learned jointly during pre-training. [8] provided among the first direct evidence that retrieval augmentation reduces hallucination specifically, rather than merely improving task accuracy. Within three years RAG had become the default architecture for deploying language models against private, current or specialist data.

The paradigm is now sufficiently mature that its limits are visible. Three findings in particular motivate this review. First, [9] demonstrated that retrieved evidence placed in the middle of a long context is substantially less likely to be used than evidence at either end, so a retrieval success can still produce a grounding failure. Second, [10] showed that even strong models rarely reject unanswerable queries and frequently follow retrieved evidence that contradicts their own correct knowledge. Third, and most consequentially, [11] constructed a large word-level annotated corpus of genuinely retrieval-augmented output and found that hallucination persists even when the relevant evidence has

been successfully retrieved. That last result undercuts the assumption on which the paradigm rests - that supplying evidence constrains generation - and it is the central problem this review organises itself around.

This paper makes four contributions. (i) It systematically reviews 35 primary studies from 2019 to 2026, with every bibliographic record resolved against publisher metadata rather than transcribed, and reports the verification procedure so the corpus is reproducible. (ii) It reconstructs the field's evolution as four distinct shifts and identifies 2023 as the inflection at which retrieval changed from a fixed pipeline stage into a decision taken during generation. (iii) It compares mitigation architectures along dimensions that are actually comparable - retrieval trigger, verification mechanism, training requirement - and argues explicitly that the cross-paper accuracy comparisons common in this literature are invalid. (iv) It mines limitations across the corpus, derives seven research gaps from their frequency distribution rather than from speculation, and proposes seven future directions and three novel research ideas.

The remainder of the paper is organised as follows. Section II describes the review methodology. Section III surveys the literature thematically. Section IV presents the comparative analysis. Section V traces research trends. Section VI derives the research gaps. Section VII proposes future directions. Section VIII concludes.

## II. RESEARCH METHODOLOGY

This review follows a structured four-stage protocol: identification, screening, verification and synthesis. The verification stage is described in detail because it departs from common practice and materially affects the reliability of what follows.

**A. Identification.** Papers were identified through three complementary routes. A seed set of architecture-defining works was drawn from recent surveys [12], [13], [14]. Forward and backward citation chaining from that seed set captured both antecedents and derivative methods. Systematic database queries were then issued against the CrossRef REST API and the arXiv API to capture peer-reviewed journal work that citation-chaining from preprints systematically under-represents. Search terms combined core concepts ("retrieval-augmented generation", "hallucination mitigation", "factuality", "grounding", "attribution") with architectural and evaluation terms. The unqualified term "RAG" was never used alone because it returns substantial noise from unrelated disciplines.

**B. Screening.** Inclusion required that a work be published between 2015 and 2026; appear in IEEE, Springer, ACM, Elsevier, MDPI, Wiley, the ACL Anthology or arXiv; and make a primary contribution to retrieval architecture, hallucination characterisation, detection, mitigation or evaluation. Works were excluded if they applied RAG to a domain without contributing a methodological advance, or if bibliographic details could not be independently resolved. The final corpus comprises 35 primary stud-

ies carried through full analysis, plus 10 additional verified works cited for context, totalling 45 references.

**C. Verification.** Every record was resolved programmatically: DOI-bearing works against the CrossRef API, and preprints against the arXiv API. Title, author list, publication year and venue were taken from the publisher's own record rather than transcribed from a citing paper or supplied by a language model. This step was not optional. During pilot work, language models asked directly for bibliographic details produced correctly formatted references to non-existent papers, and real papers with incorrect venues or years - a failure mode documented at scale in [11] and consistent with the fabricated-citation behaviour reported in the hallucination literature [15], [16]. Three papers in the corpus required a publication-year correction because the proceedings year differs from the preprint posting year. The resolution script is submitted with this review, making the corpus independently reproducible.

**D. Synthesis.** Each primary study was analysed against a fixed eight-field schema: research problem, methodology, dataset, evaluation metrics, key findings, advantages, limitations and future scope. Fields not stated in a paper were recorded as such rather than inferred. Limitations were separated into those the authors acknowledge explicitly and those inferred by this review, and were then coded into eight categories. Research gaps were derived from the resulting frequency distribution rather than generated speculatively; this follows directly from the observation that asking a generative model to "identify research gaps" produces plausible but corpus-independent suggestions.

**E. Threats to validity.** Three are material. The corpus is concentrated in English-language, Wikipedia-grounded factoid benchmarks, which limits external validity to specialist domains. A substantial minority of the most-cited works are preprints and therefore unrefereed; publication status is recorded for each. Finally, several evaluation results reported in the reviewed literature depend on language-model judges whose reliability is not independently established, which means some of the numbers synthesised here inherit an unquantified uncertainty. This is treated as a finding rather than a caveat and is developed in Section VI.

## III. LITERATURE REVIEW

The corpus organises into five themes: retrieval foundations, the limits of parametric and contextual memory, hallucination characterisation, detection and measurement, and mitigation architectures.

**A. Retrieval foundations and RAG architectures.** The paradigm rests on learned dense retrieval. [6] replaced lexical matching with a dual-encoder trained using in-batch and hard negatives, achieving large gains in top-k retrieval accuracy over BM25 with modest supervision, and became the default retriever for subsequent systems. [5] combined such a retriever with a generative reader and trained the pair end to end, establishing both the architecture and its key operational property: the knowledge

index can be replaced without retraining the model. [7] took this further by learning retrieval jointly during pre-training, making the knowledge source inspectable at the cost of substantial pre-training expense. [17] addressed evidence aggregation, encoding retrieved passages independently and letting the decoder attend jointly across them, with accuracy that scales with passage count.

Scaling followed. [18] demonstrated that retrieval over a trillion-token database allows a comparatively small model to match much larger parametric ones, decoupling knowledge capacity from parameter count. [19] supplied the infrastructure making this feasible, showing that billion-scale approximate nearest-neighbour search is practical on GPUs. A parallel line addressed deployment reality: [20] showed that simply placing retrieved passages in the prompt of a frozen model yields substantial gains without any model modification, and [21] trained a retriever against black-box model feedback, extending retrieval augmentation to systems available only through APIs.

**B. Limits of parametric and contextual memory.** [3] established that language models encode substantial relational knowledge but recall it unevenly and with high sensitivity to phrasing. [4] converted this into a design principle by relating accuracy to entity popularity, showing that retrieval helps most on long-tail entities and can actively degrade answers on popular ones - the empirical basis for adaptive rather than always-on retrieval. [9] identified an orthogonal constraint: models use evidence at the beginning and end of a long context far more reliably than evidence in the middle, producing a U-shaped performance curve. This matters because it means retrieval quality and grounding quality are separable failures; a system can retrieve correctly and still fail to use what it retrieved.

**C. Characterising hallucination.** [15] provided the field's reference taxonomy, distinguishing intrinsic unfaithfulness to a source from extrinsic unverifiable content, and tracing causes to both source-reference divergence in data and to modelling and inference. [16] updated this for instruction-tuned models, organising causes across data, training and inference stages and separating factuality from faithfulness. [22] synthesises the same territory with an emphasis on hallucination as a systemic consequence of the training objective. The theoretical position hardened subsequently: [1] argues via a computability framing that hallucination cannot be fully eliminated for any learner of this class, and [2] attributes its persistence to evaluation incentives that penalise abstention. The practical implication, which this review adopts, is that mitigation targets a bounded residual rather than zero.

**D. Detection and measurement.** [23] showed hallucination can be detected without external resources or model internals by sampling several generations and measuring their mutual consistency, on the reasoning that fabricated content is less stable across samples. [24] refined the underlying signal by clustering samples into semantic equivalence classes and computing entropy over meanings rather than token sequences, substan-

tially improving detection and demonstrating generality across models and datasets. Measurement of long-form output was addressed by [25], which decomposes generated text into atomic facts and verifies each independently, producing an interpretable proportion-supported score that has become standard. Benchmarks developed alongside: [26] constructed adversarial questions targeting common human misconceptions and found that larger models can be less truthful, ending the assumption that scale alone resolves factuality. For RAG specifically, [10] decomposed the paradigm's demands into noise robustness, negative rejection, information integration and counterfactual robustness, finding all four weak in practice. [11] supplied the supervised resource the area lacked - word-level hallucination annotation over naturally generated RAG output - and [27] addressed the annotation bottleneck by automating hallucination corpus construction.

**E. Mitigation architectures.** Mitigation strategies divide by where they intervene. Retrieval-time approaches change when and what is retrieved: [28] triggers retrieval when upcoming generation contains low-confidence tokens, and [29] interleaves retrieval with chain-of-thought so each reasoning step forms the next query, substantially improving multi-hop recall. Evidence-time approaches assess what was retrieved: [30] introduces a lightweight evaluator that classifies retrieval quality and triggers correction or web fallback, targeting the dominant RAG error source without retraining the generator. A related line intervenes mid-generation rather than before it: [31] detects low-confidence spans as they are produced and repairs them before the error propagates, on the reasoning that hallucinations compound once asserted. Generation-time approaches verify the output: [32] trains a model to emit reflection tokens that decide when to retrieve and critique whether its own output is supported, outperforming both standard RAG and larger instruction-tuned baselines on factuality and citation accuracy. [33] achieves comparable gains purely through prompting by drafting, planning verification questions, answering them independently of the draft, and revising - with the independence of the verification step essential to the effect. Post-hoc approaches operate on completed output: [34] researches and minimally revises generated text to agree with retrieved evidence while preserving the original content. Finally, structural approaches change the index itself: [35] builds an entity graph with hierarchical community summaries, enabling corpus-level questions that chunk-based retrieval cannot address, and [36] organises documents into a recursive summary tree for multi-granularity retrieval; [37] surveys this graph-structured branch as a distinct family with its own trade-offs between index construction cost and relational reasoning capability. Deployment evidence is beginning to appear: [38] compares twelve variants under one protocol on clinical vignettes, [39] reports outcomes across three industrial use cases, and [40] extends hallucination evaluation to vision-language models in the medical domain, where the consequences of unsupported output are most acute.

## IV. COMPARATIVE ANALYSIS

**A. Why a numeric leaderboard is not presented.** The systems reviewed here were evaluated with different base models, different retrievers, different corpora and different splits. [5] reports Exact Match on Natural Questions with a fine-tuned BART generator; [32] reports FActScore and citation precision on PopQA and ASQA with a purpose-trained Llama-based model; [33] reports precision on Wikidata list questions with no retrieval at all. Assembling these into a ranked table would compare experimental configurations, not methods, and would systematically favour whichever paper used the strongest base model. This review therefore compares architectures along dimensions that are genuinely commensurable and reports headline results only with the setup attached. The one study in the corpus that supports internal numeric comparison is [38], because it evaluates twelve variants under a single protocol - and its scope is a single clinical domain with 250 vignettes, which is precisely the constraint that makes its numbers meaningful.

**B. Retrieval trigger.** Four policies appear. Always-on single-shot retrieval ([5], [6], [7], [17], [21]) is simplest and remains the production default, but [4] shows it degrades answers on popular entities where the model already knows the answer. Fixed-stride retrieval [20] retrieves periodically regardless of need. Confidence-triggered retrieval [28] fires when the anticipated next sentence contains low-probability tokens, which requires token-likelihood access and conflates lexical with semantic uncertainty. Trained-policy retrieval [32] learns when to retrieve via reflection tokens, which is the most principled approach but requires training a specialised model and therefore excludes closed APIs. No study compares these policies with the retriever and generator held constant, which is the first gap identified in Section VI.

**C. Verification mechanism.** This is the sharpest architectural divide. The foundational systems ([5], [7], [17], [18], [20]) contain no verification whatsoever: retrieved evidence is provided to the generator and the output is returned unchecked. Self-critique approaches ([32], [33]) have the model assess its own output, which is cheap and model-agnostic in [33]'s prompting form but inherits the generator's own misconceptions - if the model believes something false, its verification questions will confirm it. Evidence-assessment approaches [30] evaluate the retrieved documents rather than the output, catching retrieval failure but not grounding failure. Post-hoc attribution [34] checks the finished output against freshly retrieved evidence, which is the most independent check available but can only repair locally. No system in the corpus enforces per-claim entailment between output and retrieved evidence as a precondition for returning an answer.

**D. Training requirement.** Methods requiring generator training ([5], [7], [18], [32]) achieve the strongest reported results but cannot be applied to commercial API models, which is where most deployment now occurs. Methods requiring only retriever training [21] occupy a middle ground, though [21] depends on token likelihoods that several major APIs no longer expose.

Training-free methods ([20], [28], [29], [33], [34]) are immediately deployable and consequently dominate industrial practice, at the cost of additional inference passes.

**E. The cost dimension nobody reports.** Every verification and iterative retrieval method adds inference cost. [23] and [24] require multiple sampled generations; [28] and [29] retrieve repeatedly during generation; [33] adds several verification passes; [30] adds an evaluator and potentially a web search. None of these reports latency or token overhead against the hallucination reduction achieved. A practitioner cannot therefore determine whether a method halving hallucination at five times the latency is preferable to one reducing it by a third at 1.2 times - yet the literature presents them as directly comparable alternatives. This omission is systematic across the corpus and constitutes the third research gap.

**F. Evaluation protocol.** The corpus divides between answer-correctness metrics (Exact Match, F1) that cannot distinguish a grounded answer from a lucky one, and grounding-aware metrics that can. The latter - atomic factual precision [25], reference-free component-wise faithfulness [41], and span-level annotation [11] - emerged around 2023-24 and represent the field's clearest methodological advance. However, two of the three depend on language-model judges, introducing a validity concern developed in Section VI.

## V. RESEARCH TRENDS

Reconstructing the corpus chronologically reveals four shifts rather than continuous incremental progress.

**Shift 1: knowledge leaves the parameters (2019-2020).** [3] established that parametric recall is real but unreliable. Within a year, [6], [7] and [5] had externalised knowledge into a retrievable index, making it inspectable, updatable and attributable. The significance is architectural rather than merely quantitative: knowledge became a component that could be swapped rather than a property of the weights.

**Shift 2: retrieval scales (2021-2022).** [19] made billion-scale vector search practical and [18] demonstrated that retrieval over a sufficiently large corpus substitutes for parameters, letting a smaller model match much larger ones. [17] showed evidence aggregation scales with passage count. In the same period [26] delivered a corrective finding: larger models can be less truthful, because they model the human text distribution including its misconceptions. Scale was thereby established as a solution to knowledge capacity and explicitly not a solution to factuality.

**Shift 3: retrieval becomes a decision (2023).** This is the field's inflection point. Before it, retrieval was a fixed pre-processing step; after it, retrieval is something the system decides about during generation. [4] established that retrieval should be selective; [28] triggered it on generation confidence; [29] interleaved it with reasoning so queries depend on intermediate conclusions. In the same year hallucination became measurable rather than merely describable, through consis-

cy-based detection [23] and atomic factual precision [25]. The concurrent maturation of black-box methods ([20], [21]) meant these advances applied to commercially deployed models, not only to research systems.

**Shift 4: verification enters the loop (2024-2026).** [32] and [30] added explicit assessment - of the model's own output and of the retrieved evidence respectively - and [33] showed comparable gains through prompting alone. In parallel the index acquired structure ([35], [36]), extending RAG from local lookup to corpus-level sensemaking. Evaluation matured decisively: RAGAS [41] made component-wise diagnosis possible, RGB [10] decomposed the abilities RAG demands, and RAGTruth [11] supplied word-level supervision. From 2025 the literature consolidates: [1] and [2] establish theoretical and incentive limits, peer-reviewed syntheses appear ([13], [42], [43]), retrieval extends into multi-step tool-using agents surveyed by [44], application-oriented syntheses begin organising mitigation by deployment context [45], and [38] and [39] move the evidence base from benchmarks toward deployment.

The overall trajectory is from "fetch then generate" to "decide whether to fetch, assess what was fetched, verify what was generated, and abstain when support is absent." The unresolved element is the last one: abstention remains the least implemented behaviour in the corpus, despite [10] identifying negative rejection as a core requirement and [2] arguing that evaluation actively discourages it.

## VI. RESEARCH GAPS

Research gaps were derived from the limitation frequency distribution across the 35 primary studies rather than generated speculatively. Each of the three limitations recorded per paper was coded into one of eight categories, giving 105 codings. The distribution is dominated by generalization (26 codings, 24.8%), evaluation validity (18, 17.1%), computational cost (14, 13.3%), retrieval quality (13, 12.4%) and dataset issues (11, 10.5%), with verification absence (10, 9.5%), reproducibility (9, 8.6%) and context utilisation (4, 3.8%) following. Seven gaps follow from categories appearing in four or more papers. Each is stated with the papers evidencing it and its practical significance.

### G1: No standard way to decide when retrieval should fire.

P10 shows always-on retrieval degrades head-entity answers while helping the long tail. P25 triggers on token confidence and P23 on trained reflection tokens - two incompatible signals, never compared under one protocol. P31 shows models rarely reject unanswerable queries, so they cannot recognise when retrieval has failed. Retrieval triggering is currently a per-system heuristic. Every deployment re-derives it, and no benchmark isolates trigger quality from retrieval or generation quality, so improvements cannot be attributed.

**G2: Grounding is assumed rather than verified at generation time.** P01 and P04 contain no mechanism to check that output is entailed by retrieved passages. P22 reduces but does not eliminate hallucination. P32 demonstrates hallucination persists in

RAG output even when relevant evidence was successfully retrieved - the failure is in grounding, not retrieval. The central assumption of RAG - that retrieved evidence constrains generation - is empirically false at the margin. Without per-claim entailment checking, a RAG system cannot distinguish a supported answer from a fluent unsupported one.

### G3: Verification cost is never reported alongside accuracy.

Computational cost is the most frequent limitation in this corpus. P17 and P18 need multiple samples; P25 and P26 retrieve repeatedly; P27 adds verification passes. None of these report latency or token cost against the hallucination reduction achieved. Practitioners cannot choose a technique without a cost-benefit figure. A method that halves hallucination at five times the latency is not comparable to one that reduces it by a third at 1.2x, yet the literature presents them as if they were.

**G4: Cross-paper numeric comparison is invalid but routinely performed.** Reported systems use different base models, retrievers, corpora and splits. P34 is the only study in this corpus comparing many variants under one protocol, and it covers a single clinical domain. P30 provides component-wise metrics but with an LLM judge whose stability across models is unvalidated. Without a shared protocol the field cannot establish which mitigation actually works. Apparent progress may reflect differences in base model strength rather than in method.

**G5: Evaluation increasingly depends on LLM judges whose reliability is unestablished.** P29 evaluates global query answering with LLM-judged comprehensiveness and diversity. P30's faithfulness metric is LLM-computed. P20's atomic-fact decomposition is model-generated. P19 uses a fine-tuned judge model. Hallucination evaluation is becoming circular - models are increasingly judged by models with correlated failure modes. If judge and generator share a misconception, the metric certifies the error.

**G6: Long-tail and low-resource domains remain unaddressed.** P10 identifies the long tail as where retrieval matters most, yet P02-style retrievers need supervised in-domain pairs and degrade out of domain. P34 and P35 report that domain-specific tuning is required in clinical and industrial settings. The settings where hallucination is most consequential - specialist medical, legal, industrial - are exactly those with the least training data for retrievers and the least benchmark coverage.

**G7: Theoretical impossibility results are disconnected from practical error budgets.** P15 proves hallucination cannot be eliminated; P16 argues evaluation incentives sustain it. Neither connects to an achievable residual rate for a bounded task. P35 reports residual hallucination in production requiring human oversight, but without reference to any theoretical floor. Engineering needs a target. Without a link between the impossibility results and attainable rates on restricted domains, deployment decisions about acceptable risk have no principled basis.

## VII. FUTURE RESEARCH DIRECTIONS

Each direction below addresses a gap identified in Section VI, with an assessment of feasibility given currently available components.

**Addressing G1.** A unified retrieval-triggering benchmark that holds retriever and generator fixed and varies only the trigger policy (always-on, popularity, token confidence, semantic entropy, trained reflection). \*Expected impact:\* Makes trigger quality independently measurable and comparable for the first time. \*Feasibility:\* High - composable from existing public datasets and released trigger implementations.

**Addressing G2.** Per-claim entailment gating: decompose the draft into atomic claims, check each against retrieved passages with an NLI model, and force abstention or re-retrieval on unsupported claims before the answer is returned. \*Expected impact:\* Converts grounding from an assumption into an enforced constraint, and yields a per-claim support score for the user. \*Feasibility:\* High - combines mature atomic decomposition and NLI components.

**Addressing G3.** Mandatory cost-quality reporting: a standard triple of hallucination rate, added latency and token overhead for every proposed mitigation. \*Expected impact:\* Enables rational technique selection and exposes methods whose gains are simply bought with compute. \*Feasibility:\* High - a reporting convention, not a technical advance.

**Addressing G4.** A frozen common-protocol harness fixing base model, retriever, corpus and splits, against which any mitigation can be dropped in. \*Expected impact:\* Separates method contribution from base-model strength; makes progress claims falsifiable. \*Feasibility:\* Medium - requires community coordination and sustained maintenance.

**Addressing G5.** Judge-independence auditing: evaluate every LLM-judged metric across several judge families and report inter-judge agreement as a validity statistic. \*Expected impact:\* Quantifies how much of a reported gain is an artefact of the judge, and detects correlated generator-judge failure. \*Feasibility:\* High - a protocol addition to existing frameworks.

**Addressing G6.** Retriever domain adaptation from unlabelled corpora plus small expert-verified seed sets, evaluated specifically on long-tail entities. \*Expected impact:\* Extends RAG reliability to the specialist domains where hallucination carries the highest cost. \*Feasibility:\* Medium - unsupervised adaptation is unsolved but actively progressing.

**Addressing G7.** Empirical error-budget curves: measure achievable residual hallucination as a function of domain restriction and abstention rate, linking impossibility results to deployment targets. \*Expected impact:\* Gives engineers a defensible acceptable-risk figure rather than an aspiration of zero. \*Feasibility:\* Medium - needs careful task-distribution definition.

**Novel research proposals.** Three proposals are advanced as original contributions of this review.

**Proposal 1: Entailment-Gated RAG (EG-RAG).** RAG systems emit unsupported claims even when correct evidence was retrieved (P32). No production pipeline enforces claim-level grounding before returning an answer. Existing work either critiques with the generating model itself (P23, P27) - inheriting its misconceptions - or scores factuality after the fact (P20). EG-RAG puts an independent, non-generative NLI gate inside the response path, making abstention the default when entailment fails rather than an option the model may decline to take. \*Expected impact:\* A per-claim support score returned with every answer, and a tunable precision-abstention trade-off that a deployment can set by risk tolerance.

**Proposal 2: Entropy-Triggered Adaptive Retrieval (ETAR).** Retrieval triggers use token probability (P25) or trained reflection tokens (P23). Token probability confuses lexical with semantic uncertainty; reflection tokens need a specially trained model and so exclude closed APIs. Semantic entropy has been used only for post-hoc detection (P18). Using it as the retrieval trigger inverts it from a diagnostic into a controller, and unlike reflection tokens it needs no model training, so it works on black-box models. \*Expected impact:\* Retrieval fires when the model is uncertain about meaning rather than phrasing - fewer wasted retrievals on head entities and fewer misses on the long tail.

**Proposal 3: Adversarial Judge Auditing for RAG metrics (AJA).** RAG evaluation increasingly relies on LLM judges (P29, P30) whose reliability is unestablished, risking circular evaluation where judge and generator share failure modes. Current practice validates a judge once against human labels on one dataset. AJA instead constructs adversarial cases targeting known shared misconceptions between judge and generator, and reports a judge-independence statistic as a required component of any evaluation claim. \*Expected impact:\* Turns judge reliability from an assumption into a reported measurement, and detects the specific failure where a metric certifies an error because judge and generator agree.

## VIII. CONCLUSION

This review examined 35 primary studies published between 2019 and 2026 on retrieval-augmented generation as a response to hallucination in large language models, with every bibliographic record verified against publisher metadata rather than transcribed.

Three conclusions follow. First, the field's development is better understood as four discrete architectural shifts than as incremental refinement, with the decisive transition occurring in 2023 when retrieval changed from a fixed pipeline stage into a decision made during generation. What follows from that shift - evidence assessment, self-critique, structured indexing - are consequences of treating retrieval as a control problem rather than a preprocessing one.

Second, the paradigm's central assumption is weaker than its adoption implies. Retrieval augmentation demonstrably reduces hallucination [8], but [11] shows that RAG systems still produce

unsupported content when correct evidence has been retrieved, and [9] shows retrieved evidence is used unreliably as a function of its position in the context. Supplying evidence does not, by itself, constrain generation. No system in this corpus enforces per-claim entailment between output and retrieved evidence as a precondition for answering, which is why entailment-gated generation is proposed here as the highest-value direction.

Third, the field's evaluation practice is a live threat to its own conclusions. Cross-paper numeric comparison is routinely presented despite being invalid across differing base models and splits; verification cost is essentially never reported alongside accuracy gains; and grounding-aware metrics increasingly depend on language-model judges whose reliability against correlated failure modes is unestablished. These are not peripheral concerns - they determine whether reported progress is real.

Given [1]'s formal result that hallucination cannot be eliminated and [2]'s argument that evaluation incentives sustain it, the appropriate objective is not a hallucination-free system. It is a system whose residual error rate is bounded, measured, reported alongside its cost, and made visible to whoever relies on the output. The techniques reviewed here move toward that goal unevenly: retrieval and verification have advanced substantially, while abstention - the behaviour that makes a bounded error rate safe to deploy - remains the least developed capability in the literature.

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